

Neal Kaushik Sharma

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EDUCATION

Iowa State University

Ames, Iowa

Computer Science Honors - Minors in Artificial Intelligence, Cybersecurity, and Data Science

Expected: May 2028

GPA: 3.84/4.0 - Dean's List (All Semesters)

Junior

COMPUTATIONAL SKILLS

Languages: Java, Python, C/C++, C#, PHP, TypeScript, JavaScript, SQL, Bash, LEGv8 Assembly, HTML/CSS, LaTeX

Tools and Frameworks: Spring Boot, .NET, Laravel, Django, React, Node.js, Docker, Redis, Azure, Postman

EXPERIENCE

Application Development Intern - Fareway Stores

June 2026 - August 2026

- Developed a full-stack application used across 144 Fareway stores by leadership to track and analyze thousands of responses.
- Led project planning translating business requirements through discovery, sprint planning, user stories, and business demos.
- Built the Planogram Tracker (visual layout diagrams for shelf product placement) using Laravel/PHP, React, PostgreSQL, Redis, and external APIs, designing database structures, frontend components, backend, sync workflows, and role-based access.
- Collaborated with the web team and worked on user stories through bug fixes, bash scripting, and production maintenance.

Software Engineer Intern - Tata Consultancy Services

June 2025 - August 2025

- Developed a full-stack retail management application simulating a coffee store from scratch ([GitHub](#)).
- Built the web app using C#, ASP.NET Core MVC, SQL Server, ADO.NET, jQuery, and AJAX. Implemented authentication, image uploads, encryption, centralized logging, and MVC architecture linked to SQL Server to manage inventory, orders, and products.
- Rotated with frontend, backend, database, DevOps, and cloud engineers, gaining exposure to the full software development lifecycle and Agile workflows while learning skills in API development, Bootstrap, documentation, and Azure cloud.

Cybersecurity & Software Development Assistant - ISEAGE

January 2026 - Present

- Plan and execute multiple Cyber Defense Challenges with 100+ participants each competition by developing multi-application scenarios that simulate realistic operations, including web applications, services, and infrastructure.
- Gain hands-on experience across diverse programming languages, frameworks, Linux administration, networking, and cybersecurity by building and connecting services across different Docker containers and virtual machines.
- Lead live competitions alongside cybersecurity professionals, managing infrastructure, and coordinating the event.

Research - Malicious Unlearning Attacks against Predictive Uncertainty in AI Models

August 2025 - May 2026

- Researched machine unlearning impacts on AI models - presented at the *National Conference on Undergraduate Research*.
- Implemented and compared methods for data removal from pre-trained models on image classification models.
- Applied uncertainty estimation techniques (Deep Ensembles, Monte Carlo Dropout) and metrics such as Expected Calibration Error and Brier Score to predict model behavior, confidence, accuracy, and identify vulnerabilities introduced by unlearning.

PROJECTS [nealks.netlify.app/projects]

Saint Cy's Hospital - Hospital Management Platform [[Android GitHub](#)] [[WebApp GitHub](#)]

- Android: Awarded *Best Project Overall (1st place)* among 300+ students. Developed the backend using Spring Boot to implement appointments, prescriptions, pharmacy, Firebase notifications, video calling, and a Gemini based AI health assistant.
- Web: Developed both frontend and backend for the React/Node.js application, building patient dashboards, doctor portals, pharmacy workflows, admin management, JWT authentication, MongoDB APIs, notifications, and health record features.

Space-Miner - 2D Java Game [[GitHub](#)]

- A top-down survival/adventure game built entirely from scratch in Java, under active development with plans for public release.
- Solved gameplay complexity challenges by designing modular core gameplay features, including collision detection, procedural world generation, screen culling, dynamic lighting, and A* companion AI, improving scalability and responsiveness.
- Built game engine features including minimap, UI systems, save/load functionality, sound integration, and debugging tools.

Network Attached Storage

- Built a Raspberry Pi NAS/server with OpenMediaVault, automated multi-drive backups, SMB shares, Tailscale, and Docker.
- Configured secure networking, file versioning, remote access, and backup workflows with a custom 3D-printed enclosure.